

Harsh Shrivastava

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🎓 Google Scholar 🔗 LinkedIn 🔄 GitHub

Research Interests

- Causal Inference · Causal Discovery · Causal Representation Learning
- Causal Decision Making · Bandits · Reinforcement Learning

Education

- Aug 2025 – Present** • **M.Sc. in Machine Learning**
Mohamed bin Zayed University of Artificial Intelligence (MBZUAI), Abu Dhabi, UAE
CGPA: 3.61/4.00
Advisors: Prof. Kun Zhang, Prof. Junpei Komiyama, Prof. Jin Tian
- Aug 2016 – Aug 2020** • **B.Eng. in Computer Science and Engineering**
Rustamji Institute of Technology, BSF Academy, Gwalior, India
CGPA: 8.36/10

Selected Publications

- UAI 2026** • **Fixed-Confidence Best-Arm Identification for Causal Mediation Analysis**
Harsh Shrivastava, Yuta Kawakami, Junpei Komiyama, Jin Tian.
Conference on Uncertainty in Artificial Intelligence (UAI), 2026. Accepted.
First author.
- Under Review** • **Measuring Precision in Individual Causal Effect Prediction**
Yuta Kawakami*, Harsh Shrivastava*, Jin Tian.
*Under review. (*Equal contribution.)*

Research Experience

- May 2026 – Present** • **Johns Hopkins University, USA**
Advisor: Prof. Murat Kocaoglu
Causal Reinforcement Learning. Working at the intersection of causal inference and reinforcement learning, with emphasis on causal structure, decision making, and policy learning under interventions.
- Aug 2025 – Present** • **Mohamed bin Zayed University of Artificial Intelligence (MBZUAI), UAE**
Advisors: Prof. Kun Zhang, Prof. Jin Tian, Prof. Junpei Komiyama
Fixed-Confidence Best-Arm Identification for Causal Mediation Analysis. Formulated natural direct effect maximization as a fixed-confidence best-arm identification problem. Derived instance-dependent sample-complexity lower bounds and developed an asymptotically optimal Track-and-Stop-style algorithm. Resulted in a first-author UAI 2026 publication.
Precision in Individual Causal Effect Prediction. Analyzing the evaluation of individual causal effect prediction under latent heterogeneity. Identifying limitations of PEHE and developing precision-oriented error metrics for causal effect prediction.
- Aug 2019 – Mar 2020** • **National University of Singapore, Singapore**
Advisors: Prof. Roger Zimmermann, Dr. Yifang Yin
Label-Noise Robust Audio Representation Learning. Developed a label-noise-tolerant audio representation learning approach using graph convolutional networks and multitask learning. Resulted in an ICASSP 2020 publication.
Multi-modal Learning for Enhanced Audio Tagging. Worked on a teacher-student mutual-learning framework where an audio-visual teacher guides an audio-only student for improved audio tagging. Resulted in an AAAI 2021 publication.
- Dec 2018 – Jun 2019** • **Indraprastha Institute of Information Technology (IIIT-Delhi), India**
Advisor: Prof. Rajiv Ratn Shah
Efficient Face Retrieval System Using Faded Visual Memory. Developed a human-in-the-loop face retrieval system that iteratively refines face recognition using user-provided visual clues. Resulted in an IEEE BigMM 2019 publication.
Audio Tagging Under Weak Supervision. Implemented a multi-level feature-fusion model for audio tagging from weak labels. Resulted in an ACM Multimedia 2019 publication.

Prior Publications in Audio & Multimodal ML

1. • Yin, Y., Shrivastava, H., Zhang, Y., Liu, Z., Shah, R., & Zimmermann, R.
Enhanced Audio Tagging via Multi- to Single-Modal Teacher-Student Mutual Learning.
In *Proceedings of the 35th AAAI Conference on Artificial Intelligence (AAAI)*, 2021.
2. • Shrivastava, H., Yin, Y., Shah, R., & Zimmermann, R.
Multi-task Graph Convolutional Network for Audio Tagging with Noisy Labels.
In *Proceedings of the 45th IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2020.
3. • Yin, Y., Chiou, M.-J., Zhenguang, W., Shrivastava, H., Shah, R., & Zimmermann, R.
Multi-Level Fusion Based Class-Aware Attention Model for Weakly Labeled Audio Tagging.
In *Proceedings of the 27th ACM International Conference on Multimedia (ACM MM)*, 2019.
4. • Shrivastava, H., PVNS, R., Aggarwal, K., Ayyar, M., Yin, Y., Shah, R., & Zimmermann, R.
FaceFetch: An Efficient and Scalable Face Retrieval System That Uses Your Visual Memory.
In *Proceedings of the 5th IEEE International Conference on Multimedia Big Data (BigMM)*, 2019.

Selected Industry Experience

- Apr 2024 – Jul 2025** • **Data Scientist II. Ecom Express**, India
Developed sequential prediction models for logistics planning, shipment routing, and delivery-time estimation using large-scale historical movement data.
- Jul 2023 – Mar 2024** • **Data Scientist. Farfetch**, India
Built contextual recommendation and ranking systems for search and discovery, including cold-start handling and engagement-based optimization from logged interaction data.
- Aug 2021 – Jun 2023** • **Data Scientist. Flipkart**, India
Worked on learning-to-rank and multi-objective optimization for large-scale autosuggestion systems under partial feedback. Recipient of the FK Search RnR Award (2022).
- Sep 2020 – Aug 2021** • **Data Scientist. CARS24**, India
Developed predictive models for pricing and demand estimation in online marketplaces, covering data processing, modeling, and deployment.

Skills

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| Research Areas | • Causal inference, causal discovery, bandits, reinforcement learning, statistical learning |
| Programming | • Python, SQL, PySpark |
| ML Frameworks | • PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn, NumPy, Pandas |
| Tools | • Git, Docker, GCP, Azure, AWS |
| Academic | • Research writing, paper submission, peer-review response, teaching |